Analysis of Wave Conditions at Bretagne Sud 1

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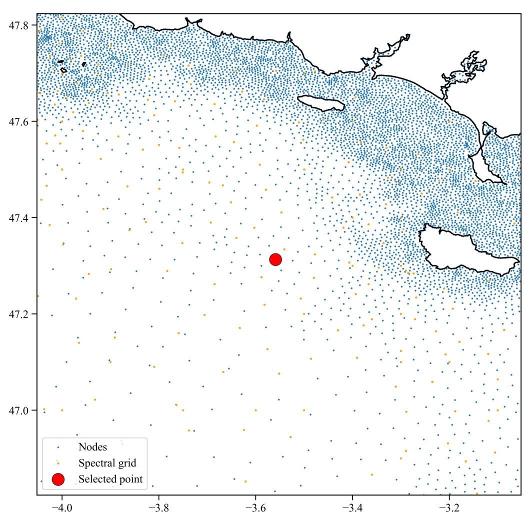
October 18, 2025

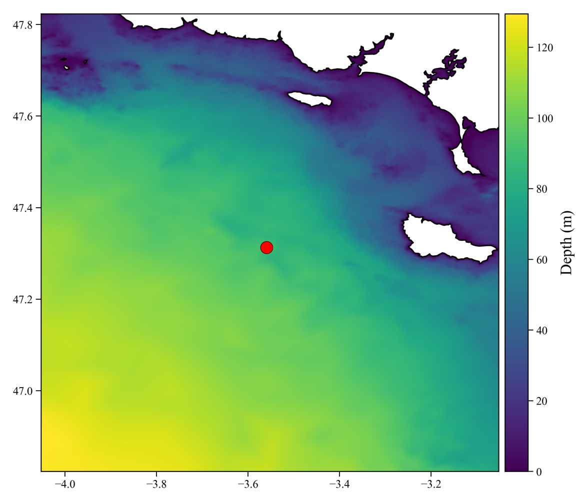
1 Introduction

This report details the analysis of wave conditions for a potential offshore wind turbine (OWT) project, covering Part I.A Mean Wave Conditions and Part I.B Extreme Wave Conditions. The objective is to characterize the wave climate at the selected study site, Bretagne Sud 1, centered near coordinates $47.32°N, -3.55°W$. Figure 1 shows the location of the analysis point within the computational grid and the local bathymetry. This analysis relies on a long-term (1994-2020) hindcast simulation dataset provided by the ResourceCode project.

Part I.A focuses on characterizing the mean wave conditions (significant wave height H_s, mean period T_{m02}, and direction), identifying the most common sea states, and assessing seasonal variability. Part I.B extends this by estimating extreme wave conditions using statistical methods. Understanding both the typical operating environment and potential extreme events is crucial for the subsequent project phases: Part II (Numerical Wave Modeling), which involves propagating waves to estimate forces, and Part III (Wave-Structure Interactions), concerning turbine movement and anchoring design.



(a) ResourceCode grid nodes



(b) Local bathymetry

Figure 1: Location of the selected ResourceCode analysis point (red dot, node 117231) near 47.32°N, 3.55°W. (a) Position within the unstructured grid (blue: all nodes; orange: spectral grid points). (b) Corresponding bathymetry (color = depth in m). The selected node is ∼1.4 km from the target coordinates.

2 Dataset

The wave data for this analysis were extracted from the ResourceCode hindcast database for the Bretagne Sud 1 site, the first commercial floating wind farm in France. The target coordinates were set to the center of the designated zone $(47.3236°N, -3.5522°W)$. The `resourcecode` tool identified the nearest available grid node (ID: 117231), located approximately 1.4 km away from these target coordinates. This offset is considered insignificant for the purpose of characterizing the general wave climate in the area.

The ResourceCode database provides long-term (1994-2020) hindcast simulations covering European North Atlantic waters ´. It was developed at Ifremer (LOPS), validated by LOPS and LCSM, and analyzed at LHEEA (Ecole Centrale Nantes). The hindcast employed the WAVEWATCH III (WW3) spectral wave model using unstructured grids with higher coastal resolution . Forcing was provided by ERA5 wind fields and surface currents derived from MARS 2D and FES2014 tidal models [1].

Hourly time series data over 27 years spanning from January 1, 1994, to December 31, 2020, were obtained for the selected node. The primary variables extracted include significant wave height, mean zero-crossing period, mean wave direction, peak period, wind components, surface current components, and water depth. Additional parameters such as wind speed, wind direction (coming-from), current speed, and current direction (going-to, derived from components) were calculated using standard conventions via the `resourcecode` utilities. Wave directions adhere to the coming-from, clockwise from North convention.

Part I
Mean Wave Conditions

3 Mean Wave Conditions and Trend Analysis (1994-2020)

This section examines the mean wave conditions at the Bretagne Sud 1 site over the 27-year period from 1994 to 2020 and assesses the presence of statistically significant long-term trends in significant wave height (H_s), mean zero crossing wave period (T_{m02}), and mean wave direction.

3.1 Methodology

Hourly data for H_s, T_{m02}, and direction (1994-2020) were aggregated into monthly means. Monthly aggregation was chosen over yearly aggregation to reduce the high variability present in hourly data while still preserving the seasonal cycle information. Two methods were applied to these monthly time series to detect potential trends:

1. **Ordinary Least Squares (OLS) Regression:** Linear regression was performed against time (in years) for each variable. For directional data, the monthly circular means were unwrapped around the overall mean direction before applying linear regression to handle the $0°/360°$ discontinuity. The slope represents the rate of change per decade.

2. **Mann-Kendall (MK) Test:** This non-parametric test was used to identify monotonic trends without assuming linearity. The test provides a p-value to assess significance and Sen's slope as an estimate of the trend magnitude.

Statistical significance for both methods was assessed using a significance level of $\alpha = 0.05$ [2].

3.2 Results

The overall mean conditions calculated over the 1994-2020 period are $H_s = 2.09$ m, $T_{m02} = 5.79$ s, and mean direction = 275.5° (coming-from). Figure 2 displays the monthly mean time series for H_s, T_{m02}, and direction, along with the fitted OLS trend lines. Table 1 summarizes the results from both trend analysis methods.

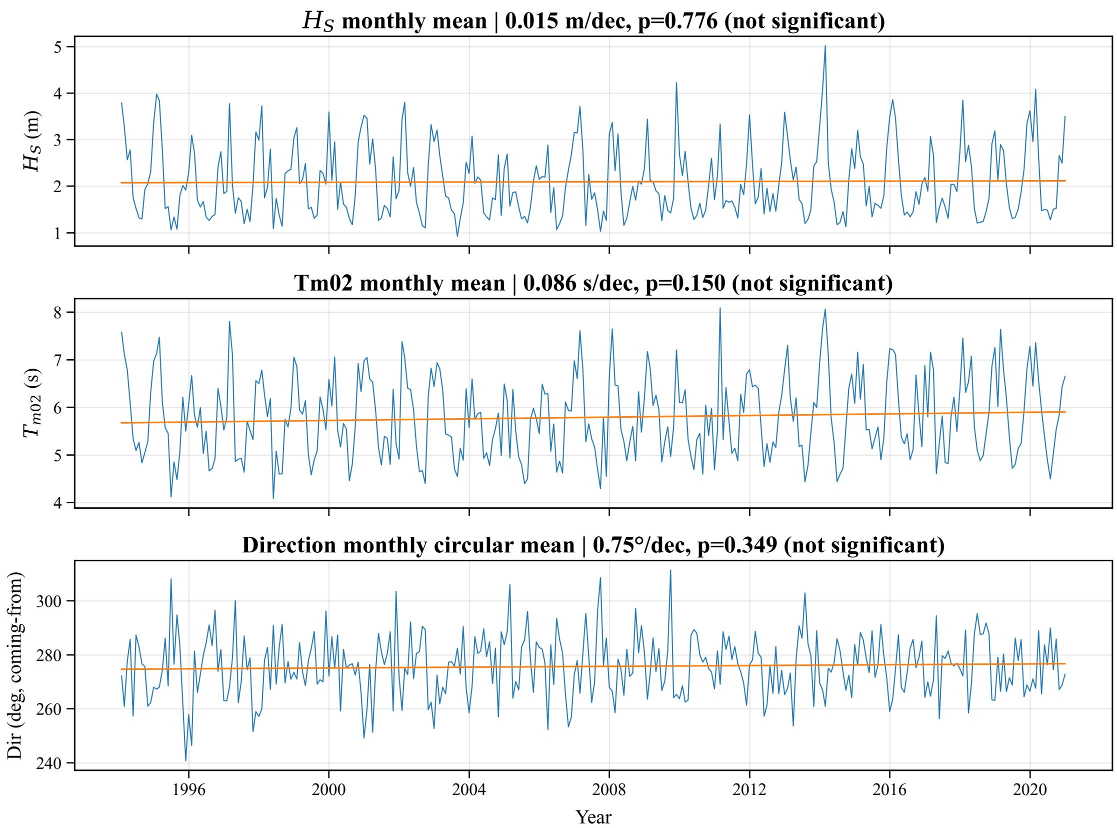


Figure 2: Monthly means of (a) significant wave height (H_s), (b) mean wave period (T_{m02}), and (c) mean wave direction (circular mean, coming-from) from 1994 to 2020. Solid lines show fitted OLS trends. Trend magnitudes and p-values from OLS are indicated in subplot titles.

Table 1: Trend Analysis Results for Monthly Mean Wave Conditions (1994-2020).

Variable	OLS Slope per decade	OLS p-value	MK Test Result (p-value)
H_s (m)	+0.015	0.776	No Trend (0.810)
T_{m02} (s)	+0.086	0.150	No Trend (0.196)
Direction (°)	+0.75	0.349	No Trend (0.441)

3.3 Discussion

The trend analyses indicate no statistically significant linear or monotonic trends ($p > 0.05$) in monthly mean significant wave height H_s, zero crossing wave period T_{m02}, or wave direction over the 1994-2020 period (Table 1). The calculated trend magnitudes are minimal compared to the observed interannual and seasonal variability (Figure 2). Extrapolating these near-zero trends suggests negligible changes in mean conditions over the expected 30-year turbine lifetime (e.g., OLS implies $\approx +0.05$ m in H_s over 30 years). While broader climate change effects remain a potential factor, this specific dataset provides no evidence of significant historical trends at this location during this period. Therefore, based on this analysis, substantial shifts in the mean wave climate due to long-term trends are not anticipated, and design considerations should prioritize characterizing the existing variability.

4 Most Common Wave Conditions

This section identifies the most frequently occurring wave conditions (H_s and T_{m02}) and the predominant wave directions at the investigated site.

4.1 Methodology

To visualize the joint distribution of significant wave height (H_s) and mean wave period (T_{m02}), a histogram was generated using the hourly data. The bin corresponding to the highest density identifies the modal (most common) condition.

The prevailing wave directions were assessed using a wave rose plot, which shows the frequency of occurrence of waves coming from different directional sectors, stacked by H_s ranges.

Wave conditions were classified into deep, transition, and shallow water regimes based on the relationship between the wave period and the mean water depth (h). Using linear wave theory approximations , the thresholds were defined based on the non-dimensional parameter $T\sqrt{g/h}$ [3]:

- Deep water: $T\sqrt{g/h} < 4$
- Shallow water: $T\sqrt{g/h}$¿25
- Transition: $4 \leq T\sqrt{g/h} \leq 25$

The mean water depth at the site was used for calculating these thresholds. The percentage of non-zero datapoints in each regime was calculated based on the T_{m02} time series.

Bin limits and outlier handling: To improve readability of the density plot, the histogram limits for H_s and T_{m02} were defined using the 0.5th and 99.5th percentiles of the respective distributions, corresponding to the central 99% of all observations. This trimming removes a few extreme outliers (often less than 0.5% of the data) that would otherwise stretch the axes and flatten the density scale, without affecting the physical interpretation of the wave climate. Consequently, the absolute extremes—although present in the full dataset—fall outside the plotting window. As a result, the upper limit of the period axis does not reach the deep-water threshold region, even though about 0.02% of values formally lie in the transition range.

4.2 Results

The mean water depth at the analysis node (117231) is 93.32 m, with a standard deviation of 1.20 m. This indicates a relatively stable depth over the period and therefore using the mean water depth is a valid simplification when calculating water regimes. Based on this mean depth, the period thresholds for wave regime classification are:

- Deep-water threshold (T_{deep}): $T_{m02} < 12.34$ s
- Shallow-water threshold ($T_{shallow}$): T_{m02}¿77.11 s

The 2D histogram of T_{m02} versus H_s is shown in Figure 3. The highest density indicates that the most common condition occurs at $T_{m02} = 3.59$ s and $H_s = 1.00$ m.

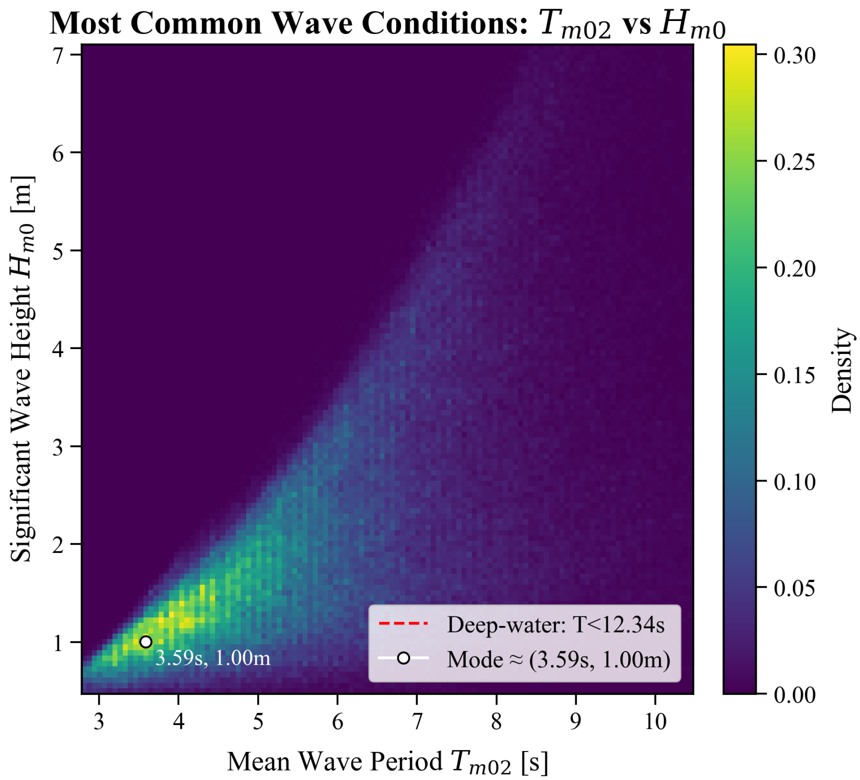


Figure 3: Density plot of mean wave period (T_{m02}) vs. significant wave height (H_s) for 1994–2020. The color indicates density of occurrence. Axes are limited to the central 99% of the data to emphasize the dominant wave climate; consequently, the upper limit does not reach the few long-period events ($\approx 0.02\%$) within the transition regime. The red dashed line marks the deep-water threshold ($T < 12.34$ s), and the white circle indicates the modal condition.

4.3 Discussion

The mean water depth at the site is 93.32 m. The analysis confirms that this location operates almost entirely under deep-water wave conditions (99.98% of the time), according to linear wave theory criteria applied to the mean period T_{m02}. As indicated in Figure 3, nearly all observed sea states fall to the left of the deep-water threshold line ($T_{m02} = 12.34$ s). Transitional conditions are extremely rare, and linear shallow-water conditions effectively never occur. This simplifies wave analysis, as deep-water approximations ($\omega^2 \approx gk$) are generally valid [3].

The most common operational condition involves relatively low waves (between $H_s \approx 0.8$ m and $H_s \approx 1.5$) with short periods (between $T_{m02} \approx 3.2$ s and $T_{m02} \approx 4.6$ s). The dominant wave direction is from the West (270°), consistent with the site's exposure to the North Atlantic. Energetic sea states (H_s¿4 m) are strongly associated with this prevailing westerly sector.

5 Seasonal Variability

This section examines the seasonal cycle of wave conditions at the Bretagne Sud 1 site, addressing the question of whether distinct seasonal patterns exist in wave height, period, and direction, and exploring the potential reasons for these patterns .

5.1 Methodology

To evaluate seasonal variability, the data for significant wave height (H_s), mean wave period (T_{m02}), and mean wave direction were grouped by calendar month. For each month (January through December), the following statistics were calculated across all years in the dataset:

- H_s and T_{m02}: The arithmetic mean and the interquartile range (IQR, difference between the 75th and 25th percentiles) were computed to represent the central tendency and typical variability.
- Direction: The circular mean and circular standard deviation were calculated to represent the average direction (coming-from) and its spread, respectively. For plotting continuity, the monthly mean directions were unwrapped relative to the overall mean direction to avoid jumps near 0°/360°.

5.2 Results

Figure 4 presents the monthly climatology for H_s, T_{m02}, and direction.

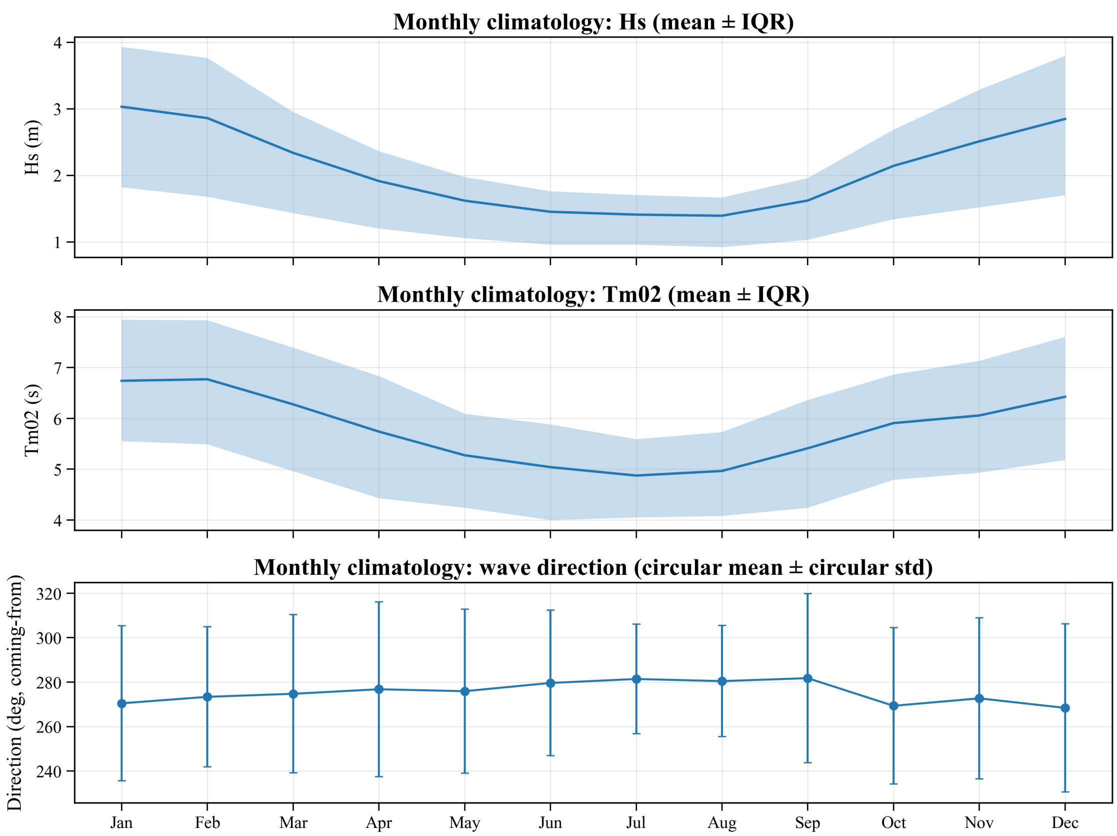


Figure 4: Monthly climatology (1994-2020) at Bretagne Sud 1. (a) Mean significant wave height (H_s) with shading indicating the interquartile range (IQR). (b) Mean wave period (T_{m02}) with IQR shading. (c) Circular mean wave direction (coming-from) with error bars representing circular standard deviation.

Key observations from the climatology are:

- Significant Wave Height (H_s): Shows a distinct seasonal cycle, peaking in winter (December/January, mean > 2.5 m) and reaching a minimum in summer (July, mean ≈ 1.5 m). The IQR (variability) is also largest in winter and smallest in summer.
- Mean Wave Period (T_{m02}): Follows a similar seasonal pattern to H_s, with longer periods in winter (peaking in January, mean ≈ 6.8 s) and shorter periods in summer (lowest in July, mean ≈ 5.0 s). Variability (IQR) is also higher in winter.

- Mean Wave Direction: Exhibits a subtle seasonal shift. The mean direction is slightly more southerly in winter (approaching 270°, West) and shifts slightly more northerly in summer (approaching 280°, WNW). The circular standard deviation (spread) remains relatively consistent throughout the year.

5.3 Discussion

Clear seasonal trends are observed in both wave height and period. H_s and T_{m02} are significantly higher and longer, respectively, during the winter months (approximately November to March) compared to the summer months (approximately May to August). This pattern is primarily driven by the seasonal variation in meteorological forcing over the North Atlantic. During boreal winter, the North Atlantic storm track is more active, generating stronger winds over larger fetches. These conditions lead to the development of larger, longer-period wind seas and swell that propagate towards the study site in the Bay of Biscay. Conversely, summer is typically characterized by weaker winds and less frequent, less intense storm systems, resulting in lower wave energy, smaller significant wave heights, and shorter mean periods. The increased variability (wider IQR) in winter reflects the greater range of conditions, from calm periods to intense storms, compared to the generally milder summer season.

A modest seasonal trend is also observed in the mean wave direction, shifting from a more westerly approach ($\approx 270°$) in winter to a slightly more west-northwesterly direction ($\approx 280°$) in summer. This shift can be attributed to the large-scale atmospheric circulation patterns. The expansion and northward shift of the Azores High during summer tend to favor more zonal (westerly) wind patterns affecting wave generation, while the increased frequency of low-pressure systems tracking into the Bay of Biscay during winter can introduce a slightly more southerly component to the incoming swell. The directional spread, however, does not show a strong seasonal signal[4].

6 Mean Wind and Current Conditions

This section evaluates the wind and current conditions at the Bretagne Sud 1 site, focusing on the mean speed and direction, as well as their seasonal variability, to complement the wave analysis .

6.1 Methodology

Wind and current speed and direction (coming-from, 0°=N, clockwise) were derived from the u and v components, respectively.

Mean conditions were calculated over the entire period and seasonally (DJF, MAM, JJA, SON, representing the binned months). Arithmetic means were used for speed, while circular means were used for direction.

Rose diagrams were generated for both wind and current to visualize the frequency distribution of speed and direction. Wind roses used 16 directional sectors (22.5°) and excluded calms ≤ 0.5 m/s. Current roses used 36 sectors (10°) and excluded calms ≤ 0.05 m/s.

6.2 Results

The overall mean wind speed is 7.22 m/s, with a mean direction of 296.6° (WNW, coming-from). The overall mean current speed is 0.096 m/s. Due to the primarily tidal, reversing nature of the currents, the overall circular mean direction (285.2°, going-to) is less physically meaningful than the directional distribution shown in the rose plots (Figure 6).

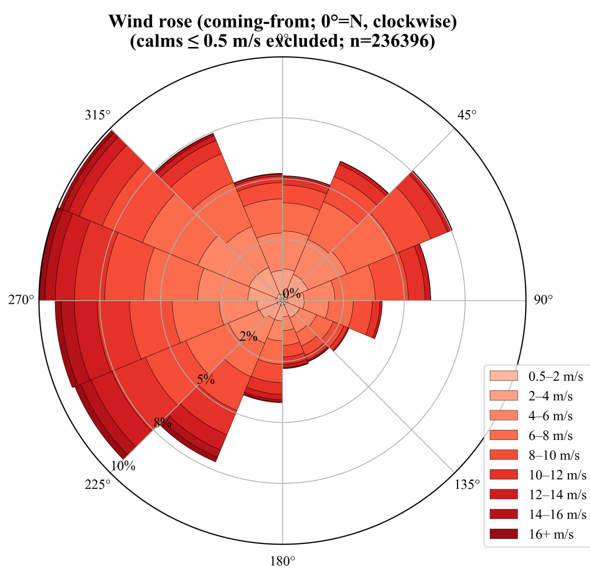
Seasonal means are presented in Table 2. Wind speeds are highest in winter (DJF mean: 8.67 m/s) and lowest in summer (JJA mean: 6.00 m/s). The mean wind direction shifts slightly, being more westerly in winter (DJF: 266.3°) and more northwesterly in spring/summer (MAM: 321.1°, JJA: 302.6°). Mean current speeds show negligible seasonal variation (≈ 0.096 m/s across all seasons).

Table 2: Seasonal Mean Wind and Current Conditions (1994-2020).

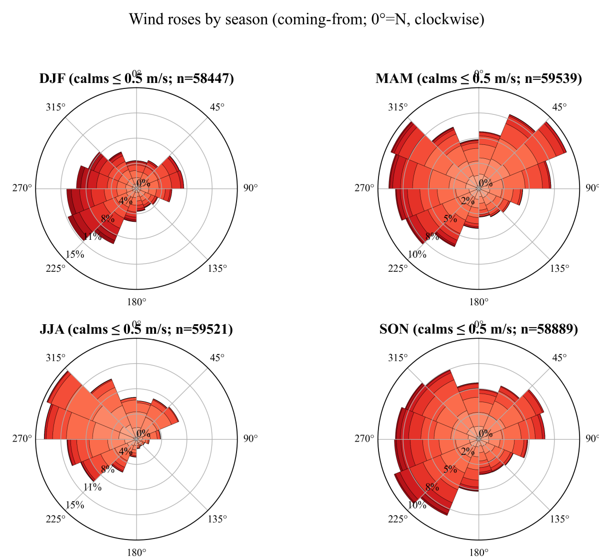
Season	Wind Speed (m/s)	Wind Dir (° from)	Current Speed (m/s)	Current Dir (° to)*
DJF	8.67	266.3	0.096	211.3
MAM	6.95	321.1	0.096	28.3
JJA	6.00	302.6	0.096	228.5
SON	7.31	287.5	0.096	19.0
Overall	7.22	296.6	0.096	285.2

*Circular mean directions for currents are less representative due to tidal reversal.

The wind rose (Figure 5a) confirms the dominant WNW wind direction, with higher speeds predominantly occurring during winter months (Figure 5b). The current rose (Figure 6a) clearly shows a bidirectional pattern, with flows primarily directed towards the NE (approx. 25°) and SW (approx. 216°). This bidirectional characteristic persists across all seasons (Figure 6b), consistent with tidally driven currents. Current speeds rarely exceed 0.2 m/s.

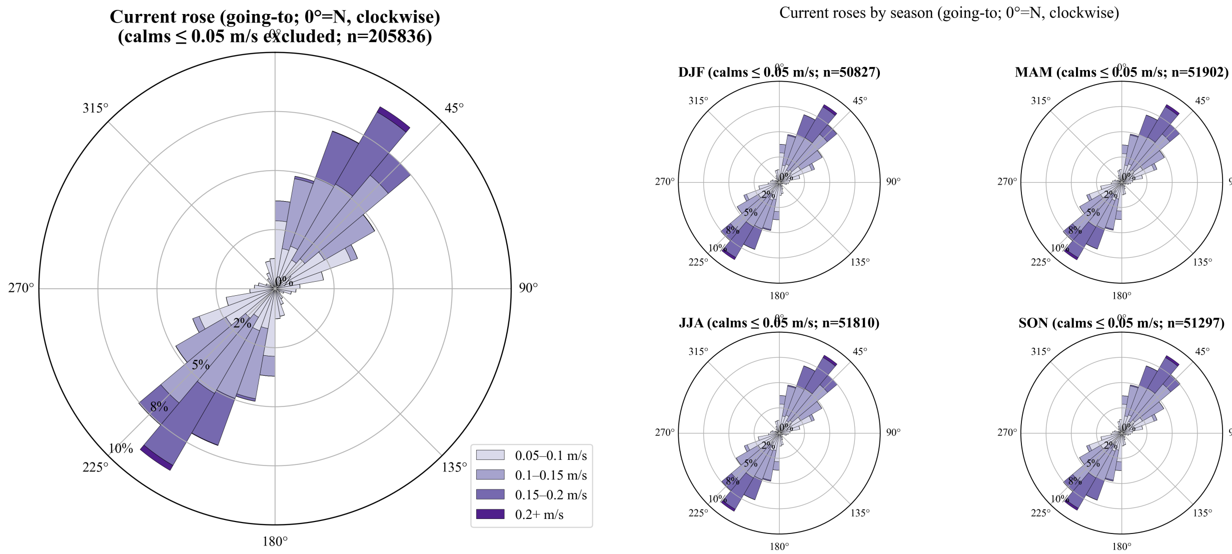


(a) Overall Wind Rose



(b) Seasonal Wind Roses

Figure 5: Wind roses (coming-from) showing frequency by direction and speed (m/s) for (a) the entire period 1994-2020 and (b) separated by season.



(a) Overall Current Rose

(b) Seasonal Current Roses

Figure 6: Current roses (going-to) showing frequency by direction and speed (m/s) for (a) the entire period 1994-2020 and (b) separated by season.

6.3 Discussion

The mean wind conditions show a clear seasonal cycle typical of the mid-latitude North Atlantic, with stronger westerly winds in winter driven by increased storm activity, and weaker, slightly more northwesterly winds in summer associated with the influence of the Azores High. Resulting in an overall mean wind speed of 7.22 m/s.

In contrast, the currents are weak (mean speed < 0.1 m/s) and exhibit strong bidirectionality along a NE-SW axis, characteristic of tidal currents. There is no significant seasonal variation observed in either the mean current speed or the dominant flow directions, confirming the dominance of tidal forcing over meteorological influences on the currents at this location. For design purposes, the bidirectional nature and peak speeds (likely below 0.3 m/s) are more relevant than the overall circular mean direction.

7 Comparison to Wave Buoy Measurements

To assess the accuracy of the ResourceCode hindcast data at the study site, a comparison was made with observational data from a nearby wave buoy for the year 2020 .

7.1 Methodology

Observational data were obtained from the Candhis network for the Belle-Île buoy (ID: 05602), located at approximately $47.31°N, -3.31°W$ in a water depth of about 45 m, for the year 2020. The buoy provides measurements including significant wave height (HM0), mean period (T02), and mean wave direction (THETAM, coming-from).

The buoy data underwent quality control (QC) prior to comparison. Records with missing values, sentinel values (e.g., ≥ 999.0), non-physical values ($H_s \leq 0$, $T_{m02} \leq 0$), extreme skewness ($|\mathrm{SKEW}| > 0.3$), or high kurtosis (KURT ge 5) were removed. If available, the 'QUALITE' flag was also used to filter out potentially erroneous data.

The buoy data (UTC) were time-matched with the hourly ResourceCode hindcast data (UTC) for the study site node (117231) during 2020. A nearest-neighbor approach was used, pairing observations within a tolerance window of ±30 minutes. ResourceCode wave direction was used directly for comparison with the buoy's THETAM.

The Root Mean Square Difference (RMSD) was calculated for H_s and T_{m02} between the paired model and observation values. For the directional comparison, the circular RMSD was calculated based on the minimal angular difference between the model and buoy directions[5].

7.2 Results

After QC and time-matching, 15,947 paired data points were available for comparison in 2020. The calculated RMSD values are presented in Table 3.

Table 3: Root Mean Square Difference (RMSD) between ResourceCode hindcast (Node 117231) and Candhis Belle-Île Buoy (05602) for 2020.

Variable	RMSD Value	N Pairs
H_s (m)	0.39	15,947
T_{m02} (s)	0.84	15,947
Direction (°)	21.2	15,947

Figure 7 shows the time series comparison for the matched data points, while Figure 8 presents scatter plots comparing the model and buoy values directly. A data gap is apparent in the buoy record during February-March 2020.

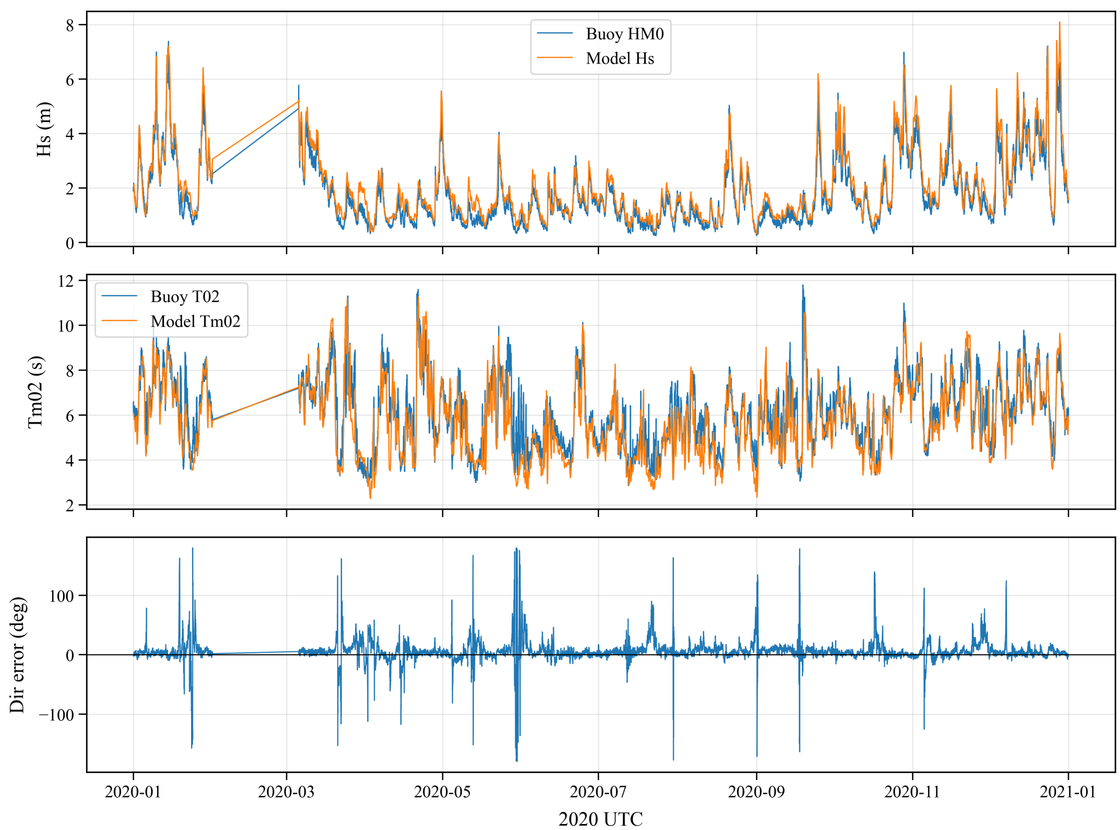


Figure 7: Time series comparison of ResourceCode hindcast (Model) and Candhis Belle-Île buoy (Buoy) data for 2020. (a) Significant wave height (H_s). (b) Mean wave period (T_{m02}). (c) Difference between model and buoy mean wave direction (coming-from).

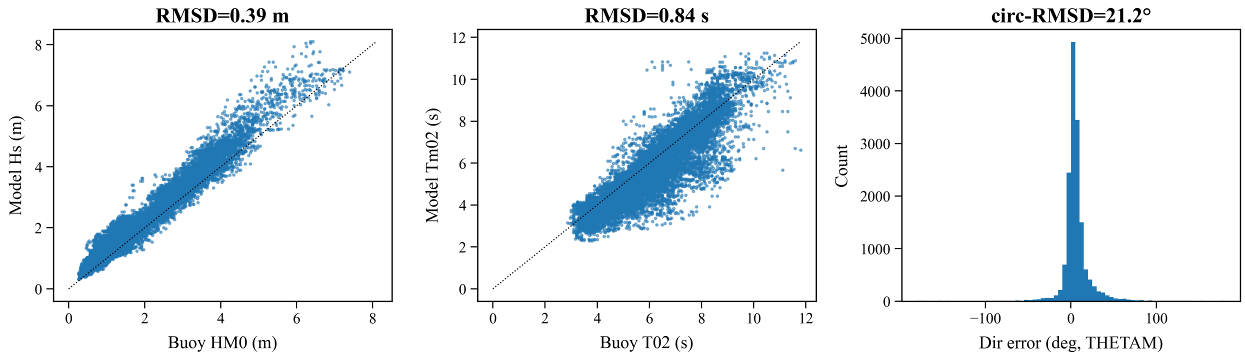


Figure 8: Scatter comparison and error distribution for ResourceCode hindcast vs Candhis Belle-Île buoy data (2020). (a) H_s scatter. (b) T_{m02} scatter. (c) Histogram of directional differences (Model - Buoy).

7.3 Discussion

The comparison between the ResourceCode hindcast and the Candhis buoy observations for 2020 yields RMSD values of 0.39 m for H_s, 0.84 s for T_{m02}, and 21.2° for mean direction. These levels of agreement are reasonable, considering the spatial separation of approximately 18 km between the model node and the buoy, and the significant difference in water depth (model: 93 m, buoy: 45 m).

The difference in water depth implies that waves, particularly longer-period swell, experience more shoaling and refraction before reaching the buoy compared to the deeper model location. This likely contributes to the observed differences, especially in wave period and direction. The H_s comparison (Figure 8a) shows generally good agreement, although the model may slightly overestimate some lower wave heights and potentially underestimate the highest peaks, which is common in point-to-point comparisons over complex bathymetry. The period comparison (Figure 8b) is also reasonable, with some scatter particularly at longer periods. The directional RMSD of 21.2° reflects challenges in accurately modeling wave direction in coastal areas subject to refraction and potentially mixed sea states. The data gap in the buoy record (February-March 2020) is noted but does not invalidate the comparison for the remainder of the year. Overall, the hindcast appears to provide a representative depiction of the wave climate, with the acknowledged differences attributable in part to the differing locations and depths[3].

Part II
Extreme Wave Conditions

8 Extreme Value Analysis

Designing marine structures like OWTs requires estimating extreme environmental conditions, particularly wave heights, to ensure structural integrity and survival during severe storms. Extreme Value Analysis (EVA) provides statistical methods to extrapolate from historical data to estimate the magnitude of events corresponding to long return periods (e.g., 1-year, 50-year). This part of the report applies two common univariate EVA methods to the significant wave height (H_s) time series from the ResourceCode dataset (1994-2020) for the Bretagne Sud 1 site: the Block Maxima (BM) method fitted with a Generalized Extreme Value (GEV) distribution, and the Peaks Over Threshold (POT) method fitted with a Generalized Pareto Distribution (GPD). The objective is to estimate the 1-year and 50-year return values for H_s and compare the results from both methods. Additionally, characteristic wave periods and directions associated with these extreme events are analyzed [6].

9 Block Maxima (BM) Analysis

9.1 Methodology

The Block Maxima (BM) method involves dividing the time series into non-overlapping blocks of equal duration and extracting the maximum value from each block . For this analysis, the 27-year (1994-2020) hourly H_s time series was divided into annual blocks, using a block size defined as 365.2425 days to account for leap years. This yielded a set of 27 annual maximum H_s values. According to extreme value theory, these block maxima should converge towards a Generalized Extreme Value (GEV) distribution. The GEV distribution was fitted to these 27 annual maxima using the `pyextremes` library. From the fitted GEV model, return levels (z_T) corresponding to specific return periods (T, e.g., 1 year, 50 years) were estimated, along with their 95% confidence intervals. Diagnostic plots (probability plot, quantile plot, return level plot, density histogram) were generated to assess the goodness-of-fit.

9.2 Results

The BM extraction yielded 27 annual maximum H_s values, ranging from 6.84 m to 11.23 m, with a mean of 8.84 m. Figure 9 shows the time series with these annual maxima highlighted.

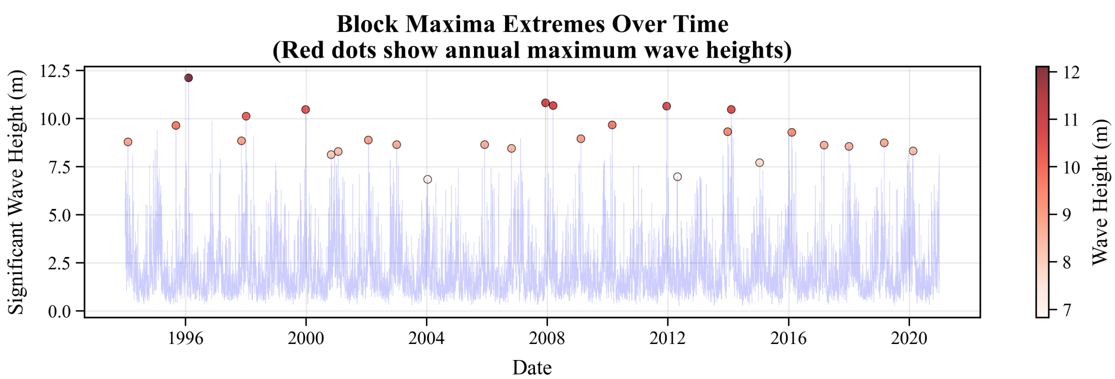


Figure 9: Hourly significant wave height (H_s) time series (1994-2020) with annual maxima identified by the Block Maxima method highlighted (colored dots).

The GEV distribution was fitted to these maxima. Diagnostic plots (Figure 10) suggest a reasonable fit, although some deviation is visible in the probability and quantile plots, potentially due to the limited number of data points (27 years).

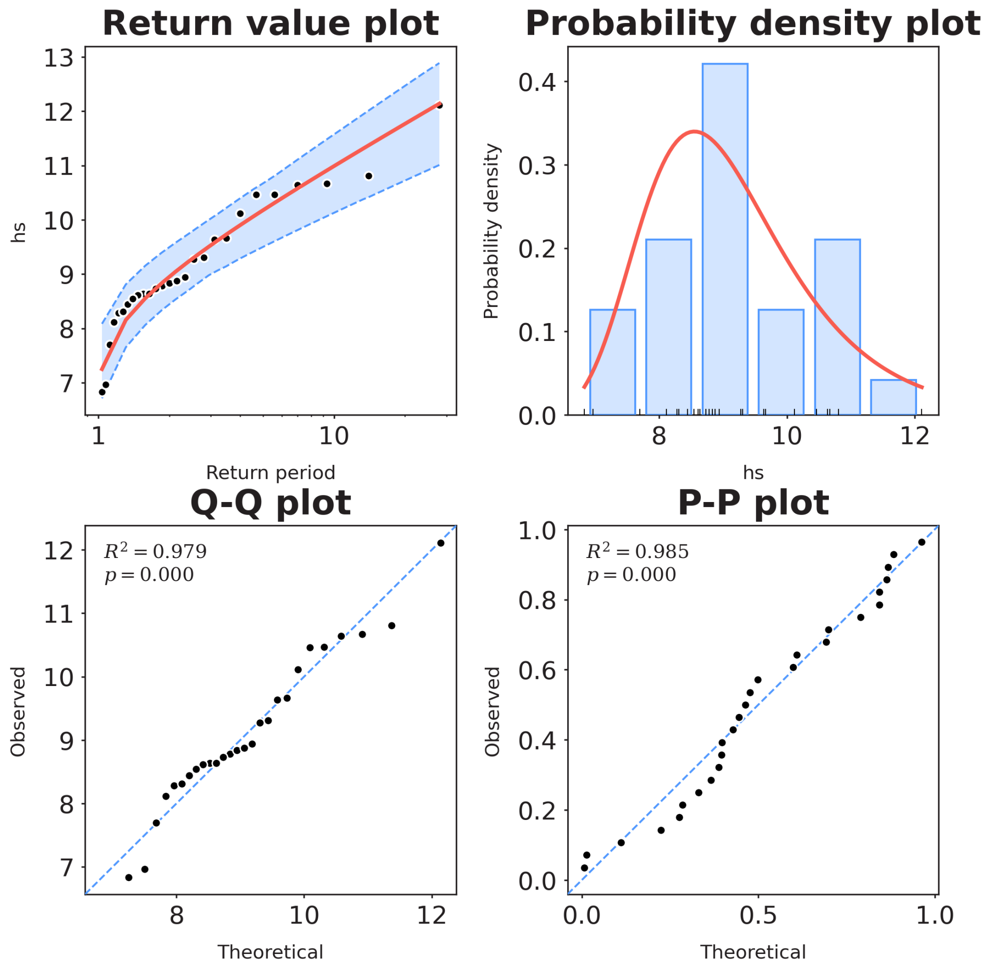


Figure 10: Diagnostic plots for the GEV fit to the annual maximum H_s values obtained via the Block Maxima method. Includes probability plot, quantile plot, return level plot (with 95% CI), and density histogram.

The estimated return values for H_s based on the fitted GEV model are summarized in Table 4 and visualized in Figure 11.

Table 4: Estimated H_s Return Values (m) using the Block Maxima (BM) Method with GEV Fit (95% Confidence Intervals).

Return Period (years)	Lower CI	Return Value	Upper CI
1.01	6.96	6.42	7.96
50	12.84	11.45	13.64

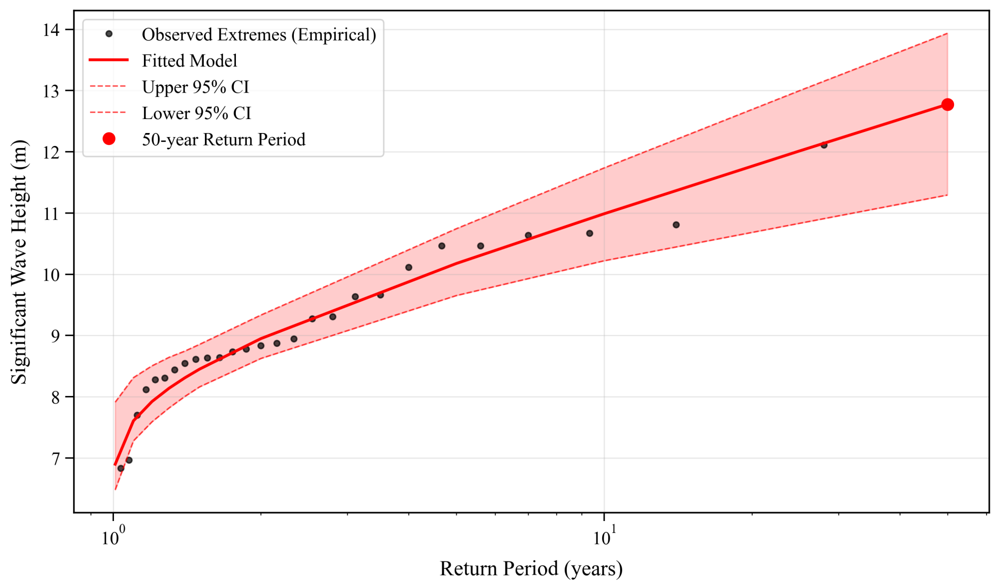


Figure 11: Return level plot for the Block Maxima (BM) analysis. Black dots represent empirical return periods of the observed annual maxima. The red line is the fitted GEV model, with dashed lines indicating the 95% confidence interval. The red circle highlights the estimated 50-year return value.

Specifically, the key estimated return values are:

- 1-year return value: 6.96 m (95% CI: [6.42, 7.96] m)
- 50-year return value: 12.84 m (95% CI: [11.45, 13.64] m)

10 Peaks Over Threshold (POT) Analysis

10.1 Methodology

The POT threshold was fixed at the 99th percentile of hourly H_s (1994–2020), following an investigation of multiple threshold-setting approaches; this choice balances sample size and extremeness by yielding enough independent exceedances while limiting contamination from non-extreme sea states. To ensure independence, a declustering window of 48 hours was applied, retaining only the highest peak within any 48-hour period above the threshold. The excesses (amounts exceeding the threshold) were fitted to a Generalized Pareto Distribution (GPD) using `pyextremes`. Return levels (z_T) for 1-year and 50-year periods were estimated with 95% confidence intervals, and diagnostic plots were generated.

10.2 Results

The 99th percentile threshold for H_s was 6.46 m. Declustering yielded 137 independent extreme events exceeding this threshold. Figure 12 indicates a good fit of the GPD model to the observed excesses.

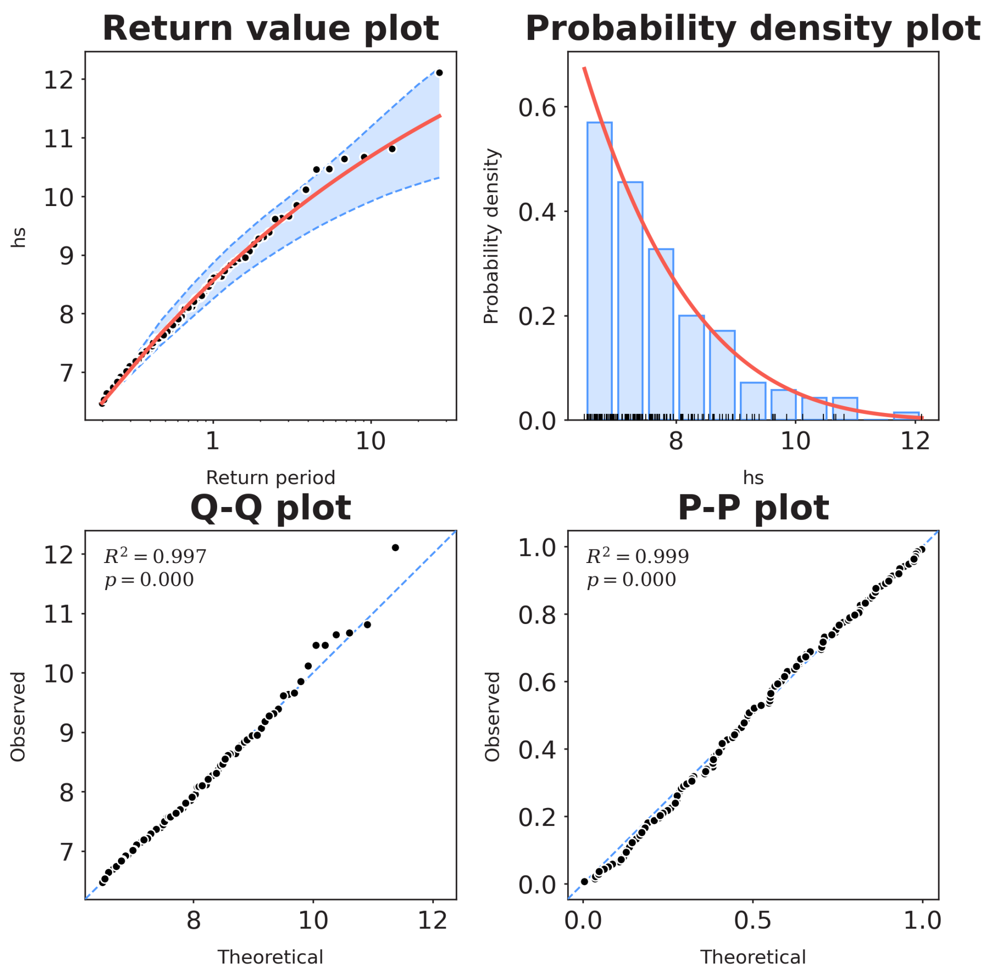


Figure 12: Diagnostic plots for the GPD fit to the H_s excesses over the 6.46 m threshold (POT method, 48hr declustering).

Estimated return values from the POT analysis are summarized in Table 5 and shown in Figure 13.

Table 5: Estimated H_s Return Values (m) using the Peaks Over Threshold (POT) Method with GPD Fit (95% Confidence Intervals).

Return Period (years)	Lower CI	Return Value	Upper CI
1.01	8.51	8.22	8.76
50	11.82	10.51	13.08

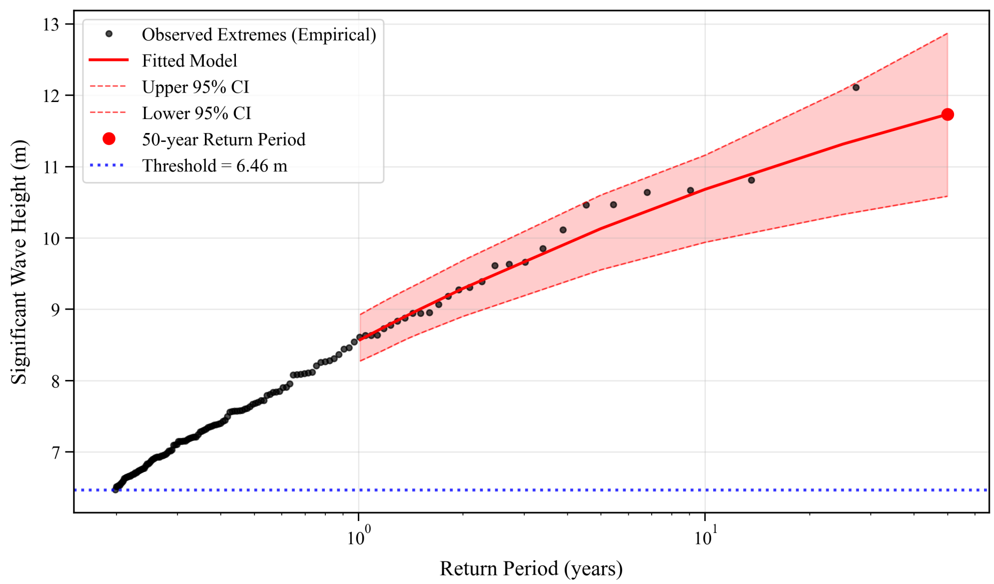


Figure 13: Return level plot for the Peaks Over Threshold (POT) analysis. Black dots: empirical return periods. Red line: fitted GPD model (dashed: 95% CI). Blue dotted line: threshold. Red circle: 50-year return value.

Specifically, the key estimated return values are:

- 1-year return value: 8.51 m (95% CI: [8.22, 8.76] m)
- 50-year return value: 11.82 m (95% CI: [10.51, 13.08] m)

11 Comparison and Discussion of BM and POT Results

This section provides a comparative analysis of the Extreme Value Analysis (EVA) results obtained using the BM and POT methods for significant wave height (H_s) at the Bretagne Sud 1 site. The comparison focuses on the estimated 1-year and 50-year return values, the associated confidence intervals, and the physical characteristics (seasonality, direction, period) of the extreme wave events identified by each method.

11.1 Comparison of Return Value Estimates

The results show a reversal in the relative ranking of the two models across return periods. For short return periods (1 year), the POT method yields a higher estimate than BM: POT 1-year = 8.51 m (95% CI: 8.22–8.76 m) versus BM 1-year = 6.96 m (95% CI: 6.42–7.96 m), a difference of 1.55 m. In contrast, for longer return periods, the BM estimate exceeds the POT estimate: BM 50-year = 12.84 m (95% CI: 11.45–13.64 m) versus POT 50-year = 11.82 m (95% CI: 10.51–13.08 m), a difference of 1.02 m. Confidence intervals overlap substantially at 50 years (11.45–13.64 m vs 10.51–13.08 m). Uncertainty patterns differ by horizon: POT is tighter at 1 year (CI width 0.54 m vs BM's 1.54 m), but slightly wider at 50 years (2.57 m vs BM's 2.19 m). The reversal implies a crossover of the fitted return-level curves at an intermediate return period.

11.2 Characteristics of Extreme Events

An analysis of the physical characteristics associated with the extreme events selected by each method provides insight into the driving conditions:

- **Seasonality:** Both methods identify extreme events that are predominantly concentrated in the winter season (December–February). The analysis of the 27 annual maxima from the BM method shows that

77.8% occurred during these months, with none recorded in summer (June–August). Visual inspection of the 137 POT events confirms a similar strong seasonal clustering in winter. This finding aligns with the seasonal climatology (Section 5), which indicated higher mean wave heights and greater variability during winter.

- **Direction:** The extreme events identified by both methods originate from a consistent directional sector. The mean direction for BM extremes is 255.9° (WSW), and for POT extremes, it is 260.4° (WSW). This WSW directionality contrasts with the overall mean wave direction of 275.5° (WNW) observed in the general climate (Section 3). Furthermore, the directional spread during these extreme events is considerably narrower (circular standard deviation of 11.6°–12.6°) compared to the overall climate (34.5°), suggesting that the most severe sea states are associated with well-defined storm systems approaching from the WSW.

- **Period:** There is a tendency for the highest-magnitude events (captured by BM) to be associated with slightly longer wave periods than the broader set of extremes identified by POT. The median mean wave period (T_{m02}) for the 27 BM events is 9.47 s, compared to 9.06 s for the 137 POT events, indicating that the largest annual wave heights are often linked to storms generating longer-period waves.

11.3 Discussion

The divergence in return value estimates arises from how the two approaches use the data. The BM method uses one annual maximum per year (27 points), making the GEV fit sensitive to a few peak events and to sampling variability within years. The POT method uses many more independent extremes (137 cluster maxima above the chosen threshold after 48 h declustering), which typically stabilizes estimation of the upper tail under the GPD model.

With the present fits, POT gives the higher estimate at short return periods (e.g., 1 year: 8.51 m vs 6.96 m for BM, a difference of 1.55 m), whereas BM becomes higher at longer return periods (50 years: 12.84 m vs 11.82 m for POT, a difference of 1.02 m). Uncertainty patterns also differ: at 1 year, the POT interval is much narrower (8.22–8.76 m) than BM (6.42–7.96 m), while at 50 years the BM interval is slightly narrower (11.45–13.64 m) than POT (10.51–13.08 m). The two fitted return-level curves therefore cross at some intermediate return period between 1 and 50 years. Consistent with the BM construction, the BM return-level curve passes close to the empirical plotting position of the sample maximum.

The consistent winter seasonality and WSW directional focus of extremes observed in both selections (BM and POT) indicate that the largest events in the dataset are concentrated in winter and approach from a relatively narrow sector. These features are intrinsic to the analyzed time series and are reflected in the diagnostics (QQ/PP/return-level plots; Figures 10 and 12). Interpretation of the crossover and the relative CI widths should be considered alongside threshold choice and declustering settings for POT, and the limited sample size for BM (27 annual maxima). Given the crossover behavior and overlapping uncertainties, neither method strictly dominates across all horizons. The choice of design value should be informed by diagnostic evidence of model fit (QQ/PP/return-level plots, stability under alternative thresholds and declustering windows for POT, and r-largest BM sensitivity), and engineering conservatism appropriate to the decision.

12 Associated Peak Period and Direction for Extreme Events

This section analyzes the peak wave periods (T_p) and mean wave directions associated with the extreme significant wave height (H_s) events identified by the Block Maxima (BM) and Peaks Over Threshold (POT) methods, as required for defining design load cases.

12.1 Methodology

The peak wave periods (T_p) and mean wave directions corresponding to the exact times of the extreme H_s events (27 for BM, 137 for POT) were extracted from the full hourly dataset.

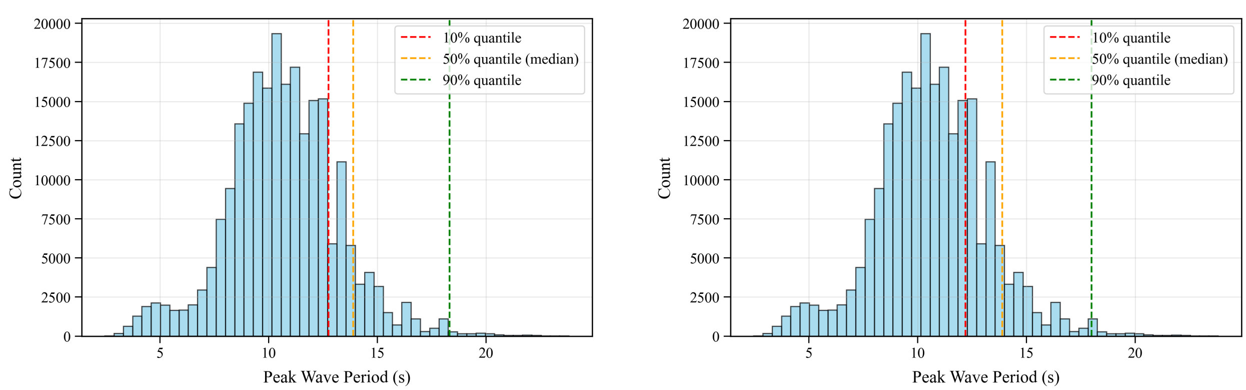
- **Period Analysis**: The 10th, 50th (median), and 90th percentiles of the associated T_p values were calculated for both the BM and POT event sets to characterize the typical range of peak periods during extreme conditions.
- **Direction Analysis**: The circular mean and circular standard deviation of the associated mean wave directions were calculated for both the BM and POT event sets. These statistics were also computed for the entire 1994-2020 dataset for comparison .

12.2 Results

Associated peak wave period (T_p) and mean direction at the time of extreme H_s events were summarized for the BM and POT selections. Statistics are reported in Table 6. Figure 14 shows the overall distribution of T_p in the dataset, with quantile lines derived from the T_p values associated with the extreme events overlaid.

Table 6: Associated peak wave period (T_p) and mean direction for extreme H_s events and for the full dataset. Directions use the coming-from convention.

Dataset	N Events	T_p Percentiles (s)			Mean Direction (°)	Circ. Std Dev (°)
		10%	50% (Median)	90%	(coming-from)	
BM Extremes	27	12.76	13.89	18.32	255.9	11.6
POT Extremes	137	12.20	13.89	17.99	260.4	12.6
All Data	236688	–	–	–	275.5	34.5



(a) BM extremes' quantiles

(b) POT extremes' quantiles

Figure 14: Overall distribution of peak wave period (T_p) from the entire dataset (1994-2020). Dashed lines mark the 10%, 50% (median), and 90% quantiles calculated from the T_p values associated specifically with the extreme H_s events identified by (a) the BM method and (b) the POT method.

The analysis indicates that extreme H_s events are typically associated with median peak periods (T_p) of 13.89 s (identical median for both BM and POT extremes). The central range (10th to 90th percentile) spans approximately 12.2 s to 18.3 s. The mean direction during these events remains consistently from the WSW (256°-260°), with a low circular standard deviation (11.6°-12.6°).

12.3 Discussion

Table 6 summarizes the key peak period (T_p) and direction characteristics associated with extreme wave heights.

Comparing the direction statistics confirms that the overall mean wave direction (275.5°, WNW) is not representative of conditions during extreme events (mean direction 256°-260°, WSW)

The associated peak periods (T_p median $\approx$ 13.9 s, typically ranging from $\approx$ 12 s to $\approx$ 18 s) are characteristic of swell generated by distant and/or intense storms, consistent with the high H_s values observed. These long periods are critical inputs for calculating wave loads and dynamic responses of the OWT structure. The calculated quantiles provide a relevant range of peak periods to consider in conjunction with the 50-year H_s for design sensitivity studies.

13 Comparison with Multivariate Environmental Contour

To provide a broader perspective on extreme conditions, a multivariate analysis was performed to estimate the 50-year environmental contour for significant wave height (H_s) and peak period (T_p). This contour represents the combinations of (H_s, T_p) that have a joint probability of being exceeded corresponding to a 50-year return period.

13.1 Methodology

The 50-year environmental contour was calculated using the Principal Component Analysis (PCA) based Inverse First-Order Reliability Method (I-FORM), as implemented in the `MHKiT` python toolbox . This method transforms the correlated (H_s, T_p) data into uncorrelated principal components, calculates the contour in this transformed space based on chosen marginal probability distributions, and then transforms the contour back into the original (H_s, T_p) space

The univariate 50-year H_s estimates from the BM (12.84 m) and POT (11.82 m) analyses were considered. To plot these points on the contour graph, they were paired with the median peak period (T_p) associated with the extreme events identified by each respective method (BM median $T_p \approx 13.9$ s, POT median $T_p \approx 13.9$ s).

13.2 Results

Figure 15 displays the calculated 50-year environmental contour plotted over the scatter of the hourly (T_p, H_s) data.

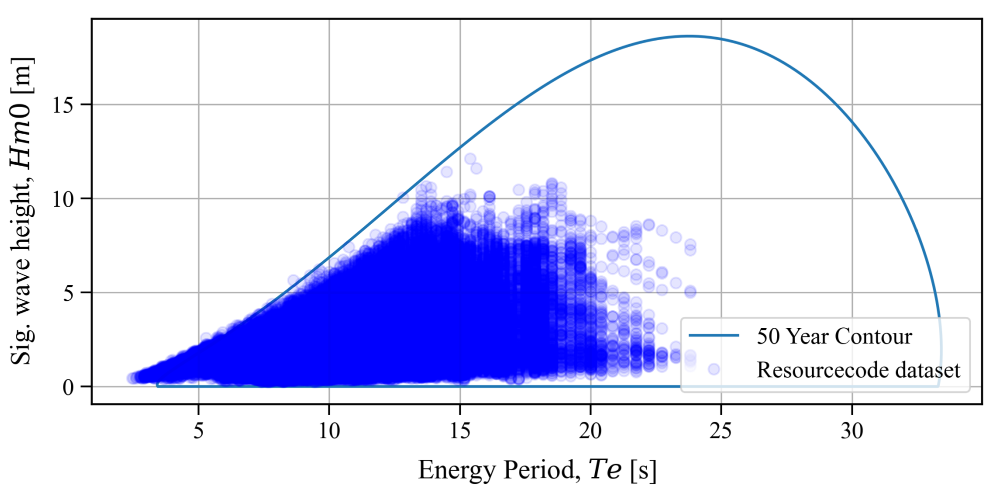


Figure 15: 50-year environmental contour for significant wave height (H_s) and peak period (T_p) using the PCA I-FORM method. The background scatter shows the hourly data points.

The contour defines a boundary encompassing various combinations of high H_s and associated T_p values expected for a 50-year joint exceedance probability. Visual inspection of the contour (Figure 15) shows that the maximum significant wave height reached along this boundary is considerably higher than the univariate 50-year estimates. The peak H_s on the contour appears to be approximately 18-19 m, occurring at a very long peak period (T_p) of roughly 23-24 s. The univariate points, representing the 50-year H_s (12.84 m and 11.82 m) paired with the median associated T_p (around 13.9 s), fall significantly below the maximum H_s point on the contour, although they lie reasonably close to the contour boundary in the region corresponding to more frequent peak periods.

13.3 Discussion

The environmental contour provides a multivariate perspective on 50-year extreme conditions, representing the joint (H_s, T_p) pairs corresponding to this return period

Comparing the results, the maximum significant wave height indicated by the contour ($\approx$ 18-19 m) is substantially higher than the 50-year H_s values estimated from the univariate BM and POT methods. Therefore, in terms of the absolute maximum predicted H_s, the multivariate and univariate results are not similar.

This difference arises because the analyses address slightly different statistical questions. The univariate EVA estimates the H_s value that is exceeded, on average, once every 50 years, regardless of the accompanying T_p. The environmental contour identifies the boundary of (H_s, T_p) pairs that are jointly exceeded with a probability corresponding to a 50-year return period. The absolute maximum H_s on the contour often occurs at the tail end of the T_p distribution (very long periods, in this case $\approx$ 24 s), representing a rarer combination than the conditions typically associated with the univariate 50-year H_s (where median $T_p \approx$ 14 s).

While the maximum H_s differs, the contour is still valuable. It confirms that the univariate 50-year H_s estimates (paired with their typical associated periods) lie near the 50-year joint probability boundary for that period range. More importantly, it provides the full range of extreme (H_s, T_p) combinations needed for comprehensive structural design, including potentially critical scenarios involving very long periods, even if the absolute maximum H_s derived from the univariate analysis is lower than the contour's peak.

Conclusion

Analysis of the 1994-2020 ResourceCode hindcast for the Bretagne Sud 1 site (mean depth 93.32 m) confirms it operates almost exclusively under deep-water wave conditions (>99.9% based on $T\sqrt{g/h}$) (Section 4). The average wave climate is characterized by $\overline{H_s} = 2.09$ m and $\overline{T_{m02}} = 5.79$ s, with prevailing waves arriving from the W-WNW sector (mean 275.5°) (Sections 3, 6). A distinct seasonal cycle exists, featuring significantly higher and more variable wave heights and periods in winter compared to summer (Section 5). No statistically significant long-term trends in mean H_s, T_{m02}, or direction were detected over the 27-year period (Section 3).

Extreme Value Analysis (EVA) using both Block Maxima (BM/GEV) and Peaks Over Threshold (POT/GPD) methods yielded differing return value estimates, showing a reversal in relative ranking across return periods (Section 8). For short return periods (1 year), the POT method yields a higher estimate than BM (POT 1-year = 8.51 m vs BM 1-year = 6.96 m, a difference of 1.55 m). In contrast, for longer return periods, the BM estimate exceeds the POT estimate (BM 50-year = 12.84 m vs POT 50-year = 11.82 m, a difference of 1.02 m) (Section 11). Confidence intervals overlap substantially at 50 years. Uncertainty patterns also differ: the POT interval is tighter at 1 year, while the BM interval is slightly tighter at 50 years (Section 11). This reversal implies a crossover of the fitted return-level curves at an intermediate return period. Given this crossover behavior and overlapping uncertainties at the 50-year horizon, neither method strictly dominates. The choice of design value should be informed by diagnostic evidence (Figures 10 & 12) and appropriate engineering conservatism.

Crucially, extreme H_s events exhibit distinct characteristics compared to the mean climate: they are predominantly concentrated in winter and consistently approach from the WSW sector (mean 255-260°), differing significantly from the overall mean direction and showing much lower directional spread (Section 11, 12). The associated peak periods (T_p) have a median of 13.9 s and typically range between approximately 12 s and 18 s (Table 6). The optional multivariate environmental contour analysis suggests potentially higher H_s values (≈18-19 m) might occur jointly with very long periods (≈24 s) within a 50-year joint probability framework, reinforcing the need to consider a range of extreme conditions (Section 13).

Validation against 2020 Belle-Île buoy data showed reasonable hindcast performance (e.g., H_s RMSD = 0.39 m) (Section 7). Uncertainties remain regarding climate stationarity assumptions and EVA parameter choices.

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